



ST BENEDICT'S

SUBJECT	PHYSICS - FINAL EXAM - MEMO
GRADE	11
EXAMINER	Mr Hilder
MODERATOR	Mrs Govender
DATE	09 November 2024
MARKS	135
DURATION	120 minutes

%

MEMO

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	Total
Question Total	15	26	17	16	11	16	14	12	8	135

Comment:

QUESTION 1

14 MARKS

1.1 The rate of change of velocity. ✓

(1)

1.2 Given: $\Delta s = 25 \text{ m}$, $\Delta t = 5 \text{ s}$, $u = 0 \text{ m.s}^{-1}$.

(3)

$$s = ut + \frac{1}{2}at^2 \quad \checkmark$$

$$25 = 0(5) + \frac{1}{2}a(5^2) \quad \checkmark$$

$$25 = 12.5a$$

$$a = \frac{25}{12.5} = \boxed{2 \text{ m.s}^{-2}} \quad \checkmark$$

→ correct equation → substitution of values. → final answer.

1.3 Velocity → 5 to 10 s

(2)

Velocity when car is done accelerating at $a = 2 \text{ m.s}^{-2}$ for $\Delta t = 5 \text{ s}$ $v = u + at \quad \checkmark$ $v = 0 + (2)(5)$ $v = \boxed{10 \text{ m.s}^{-1} \text{ east}} \quad \checkmark$	OR	v will be = to gradient $v = \frac{\Delta x}{\Delta t} = \frac{y_2 - y_1}{x_2 - x_1} \quad \checkmark$ $v = \frac{75 - 25}{10 - 5} = \frac{50}{5}$ $v = \boxed{10 \text{ m.s}^{-1} \text{ east}} \quad \checkmark$
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→ correct equation. → correct answer (including direction)

1.4 Describe motion

(4)

10-15 seconds:

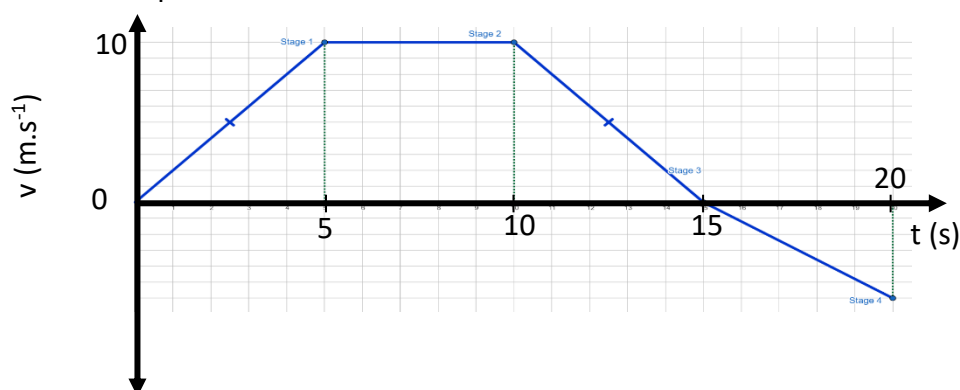
negative acceleration (west), ✓ decreasing velocity east ✓

15-20 seconds

negative acceleration (west) ✓, increasing velocity west ✓

1.5 Graph

(4)

→ Gradient of 0-5s and 10-15s equal but opposite ($m = -m$) ✓→ Constant velocity from 5 to 10 seconds. (10 m.s^{-1}) ✓

→ Gradient of 10-15s steeper than 15-20s, and goes into negative. ✓

→ Award marks for the correct shapes, **labels** ✓, and clear indication of each phase.

QUESTION 2

26 MARKS

[up is positive]2.1 Time taken: [$v_i = 12 \text{ m.s}^{-1}$; $g = -9,8 \text{ m.s}^{-2}$; $v_f = 0 \text{ m.s}^{-1}$; $t = ?$] (2)

$$v_f = v_i + at$$

$$0 = 12 + (-9,8)t \checkmark$$

$$t = \boxed{1,22 \text{ s}} \checkmark$$

→ signs differ in substitution → final answer

2.2 Highest Point (3)

$$v_f^2 = v_i^2 + 2g\Delta y$$

$$0^2 = 12^2 + 2(-9,8)(y_f - 45) \checkmark$$

$$\frac{-144}{-19,6} = y_f - 45$$

$$y_f = 7,34 + 45 = \boxed{52,32 \text{ m}} \checkmark$$

→ equation, signs, substitution → adding initial position → final answer

2.3 Difference in time. (4)

$\Delta y = u_A t_A + \frac{1}{2} g t_A^2 \checkmark$ $-45 = 12t + \frac{1}{2} (-9,8)(t^2) \checkmark$ $0 = -4,9t^2 + 12t + 45$ (quadratic equation) $t = 4,49 \text{ s} \checkmark \checkmark$	OR	$v_f^2 = v_i^2 + 2g\Delta y$ $v_f^2 = 12^2 + 2(-9,8)(0 - 45) \checkmark$ $v_f = \sqrt{1026} = -32,03 \text{ m.s}^{-1} \checkmark$ $v_f = v_i + g\Delta t$ $-32,03 = 12 + (-9,8)(t) \checkmark$ $t = \boxed{4,49 \text{ s}} \checkmark$
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2.4 BOUNCING GRAPH

2.4.1 THE SOLID LINE ✓ (1)

2.4.2 BALL A ✓ ✓ (2)

2.4.3 BALL B ✓

Ball B travels for less time than Ball A. ✓

Both ball's have the same displacement and ✓

 $v_{avg} \propto \frac{1}{t}$ when Δx is constant. (3)

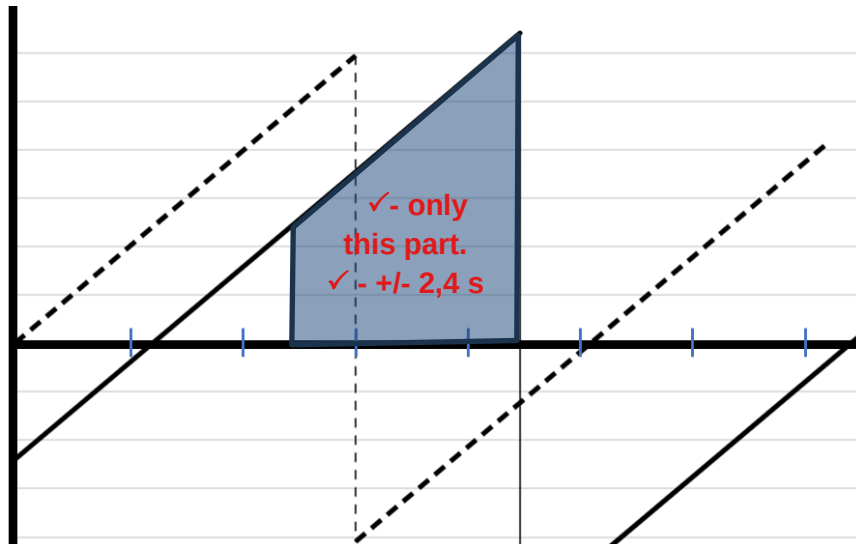
2.4.4 BALL B ✓

It experiences a greater percentage change in velocity during contact with the floor. (while bouncing) ✓

(2)

2.5

(2)



2.6 Bounce Velocity

(3)

$$v_f^2 = v_i^2 + 2a\Delta y \checkmark$$

$$0 \checkmark - v_i^2 = 2(-9,8)(45 \checkmark)$$

$$-v_i^2 = -882$$

$$v_i = \sqrt{882} = \boxed{29,7 \text{ m} \cdot \text{s}^{-1} \text{ upwards}}$$

2.7 Net acceleration

(3)

$$v_f = v_i + g\Delta t$$

$$v_f = 12 + (-9,8)(4,49 \text{ s})$$

$$v_f = \boxed{-32,03 \text{ m} \cdot \text{s}^{-1} \checkmark}$$

$$v_{\text{bounce}} = v_{\text{land}} + at$$

$$29,7 = -32,03 + a(0,13) \checkmark (\text{coe: 2.6})$$

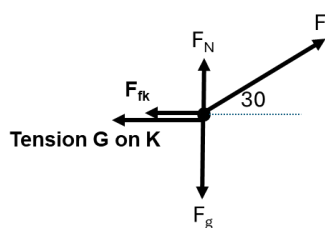
$$61,73 = 0,13a$$

$$a = \boxed{474,85 \text{ m} \cdot \text{s}^{-2} \text{ upwards} \checkmark}$$

QUESTION 3

17 MARKS

3.1 Free-body

→ F_N smaller than F_g ✓→ F_{fk} and F_T parallel ✓

→ Arrows and Dots ✓

→ Labels ✓

(4)

3.2 Normal force

(3)

$$F_N = F_g - F_y$$

$$F_N = mg - F \sin \theta \quad \checkmark$$

$$F_N = (5)(9,8) - 40 \sin(30^\circ) \quad \checkmark$$

$$F_N = 49 - 20 = \boxed{29 \text{ N upwards}} \quad \checkmark$$

Coefficient of kinetic friction

(2)

$$F_N = \mu_k F_N$$

$$5,8 = \mu_k (29) \quad \checkmark \text{ (coe: 3.2)}$$

$$\mu_k = \boxed{0,2} \quad \checkmark$$

3.3 The normal force would increase ✓,

Smaller θ , will lead to a decrease in F_y , thus less upward force actin against F_g ✓

(2)

3.4 When a net force, F_{net} , is applied to an object of mass, m , it accelerates in the direction of the net force. ✓ The acceleration, a , is directly proportional to the net force and inversely proportional to the mass. ✓

(2)

3.5 Tension

(4)

$F_{\text{net}_K} = ma$ $F_x + F_{T_{GK}} + F_{fk_K} = ma$ $F \cos \theta + F_{T_{GK}} + F_{fk_A} = ma$ $40 \cos(30) + F_{T_{GK}} + (-5,8) = 5a$ $F_{T_{GK}} = \boxed{5a - 28,84} \quad \checkmark$	$F_{\text{net}_G} = ma$ $F_{T_{KG}} + F_{fk_A} = ma$ $F_{T_{KG}} + (-2,5) = 3a$ $F_{T_{KG}} = 3a + 2,5 \quad \checkmark$
$-F_{T_{GK}} = F_{T_{KG}} \quad \checkmark$ $-(\boxed{5a - 28,84}) = 3a + 2,5$ $-8a = -26,34$ $a = \boxed{3,29 \text{ m} \cdot \text{s}^{-2}}$ $F_{T_{GK}} = 5a - 28,84$ $F_{T_{GK}} = 5(\boxed{3,29}) - 28,84$ $F_{T_{GK}} = \boxed{-12,39 \text{ N}} \quad \checkmark$	

QUESTION 4

17 MARKS

- 4.1 Maximum static friction is the largest amount of friction that an object can experience before it starts moving✓.

Whereas static friction is present when a force is applied but object remains stationary.✓ (2)

- 4.2 (3)

$$F_{fs} = F_{g\parallel} \checkmark = mg \sin \theta$$

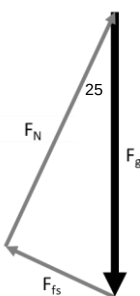
$$F_{fs} = (8)(9,8) \sin(25) \checkmark$$

$$F_{fs} = \boxed{33,13 \text{ N}} \checkmark$$

- 4.3 Vector diagram (2)

→ accurate diagram.✓

→ labels and angle✓



- 4.4 Pulley

- 4.4.1 Net force

$$F_{net} = ma$$

$$F_T + F_{g\parallel} + F_{fk} = ma \checkmark$$

$$F_T + (-33,13 \checkmark) + (-19,75) = 8(2,6)$$

$$F_T = \boxed{73,68 \text{ N}} \checkmark$$

- 4.4.2 Mass (3)

$$F_{net} = ma$$

$$F_g + F_T = ma \checkmark$$

$$9,8m + (-73,68) = 2,6m \checkmark$$

$$7,2m = 73,68$$

$$m = \boxed{10,23 \text{ kg}} \checkmark$$

- 4.5 REMAIN THE SAME✓ (1)

- 4.6 INCREASE✓

The normal force would increase due to the lower θ .✓

Friction is directly proportional to F_N . ($F_{fk} = \mu_k F_N$)✓

(if $F_{g\parallel}$ decreases, then F_f will decrease) (3)

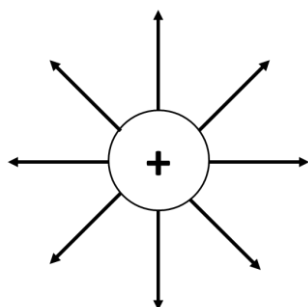
QUESTION 5 ZN**15 MARKS**

5.1 The force per unit positive charge.✓ (A.O.N)

(1)

5.2 Field pattern

(2)



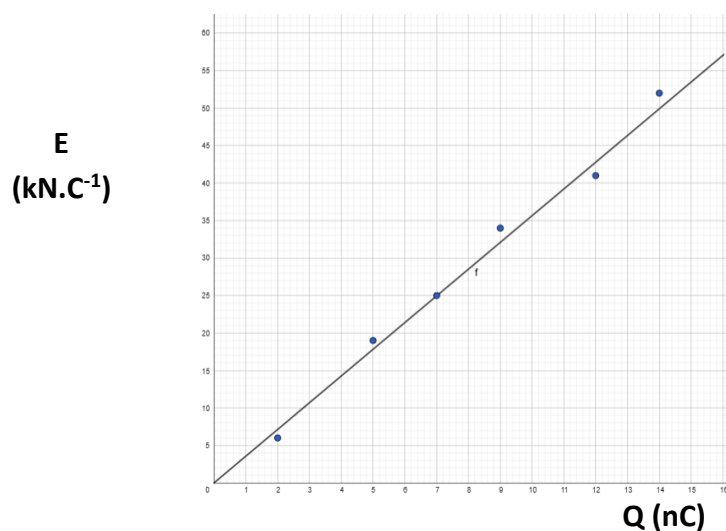
(evenly spaced and formatted✓ and arrow pointing out)✓

5.3 The charge on the sphere✓✓

(2)

5.4 Graph

(5)



y-axis and x-axis title and unit✓

scale must be in sensible multiples✓

Plotted points - check (5,19) & (12,41)✓✓

Line of best fit (with a ruler)✓

5.5 Gradient

(4)

$$m = \frac{\Delta y}{\Delta x} = \frac{\text{values from y axis}}{\text{values from x axis}} \quad \checkmark \checkmark$$

$$m = 3,67 \times 10^{12} \quad [\text{accept: } 3,3 \times 10^{12} \rightarrow 4,04 \times 10^{12}] \quad \checkmark N.C^{-2} \checkmark$$

(minus 1 if points not shown on graph)

5.6 Distance

(2)

From gradient = $\frac{E}{Q}$ $E = \frac{kQ}{r^2}$ $\therefore \text{gradient} = \frac{k}{r^2} \quad \checkmark$

$$3,67 \times 10^{12} = \frac{9 \times 10^9}{r^2} \quad (C.O.E - \text{from: 5.5})$$

$$r = 0,050 \text{ m} \quad \checkmark$$

QUESTION 6**11 MARKS**

- 6.1 Every particle with mass in the universe attracts every other particle with a force ✓ which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres ✓ (2)

- 6.2 Field strength (3)

$$g = \frac{Gm_E}{r^2}$$

$$g = \frac{(6,7 \times 10^{-11})(5,7 \times 10^{24})}{(3 \times 10^8)^2} \checkmark$$

$$g = 4,24 \times 10^{-3} \text{ N.kg}^{-1} \checkmark \text{ towards Earth} \checkmark$$

- 6.3 Net field (4)

$$g_{net} = g_E - \left(\frac{Gm_m}{r_{sm}^2} \right)$$

$$g_{net} = 4,2 \times 10^{-3} \checkmark - \frac{(6,7 \times 10^{-11})(7,35 \times 10^{22})}{(3,84 \times 10^8 - 3 \times 10^8 \checkmark)^2} \text{ (coe: 6.2)}$$

$$g_{net} = 4,24 \times 10^{-3} - 6,98 \times 10^{-4} \checkmark$$

$$g_{net} = 3,542 \times 10^{-3} \text{ N.kg}^{-1} \text{ towards Earth} \checkmark$$

- 6.4 Weight (2)

$$F_{gs} = mg \checkmark$$

$$F_{gs} = (1500)(3,542 \times 10^{-3}) \text{ [coe 5.3]}$$

$$F_{gs} = 5,31 \text{ N towards Earth} \checkmark$$

QUESTION 7

11 MARKS

- 7.1 It is the property of a substance and its ability to prevent the flow of current ✓✓
(or the measurement of the preventions of electron mobility) (2)

- 7.2 Voltmeter (2)

$$V = IR$$

$$V = (0.82)(7) \checkmark$$

$$V = \boxed{5,74 \text{ V}} \checkmark$$

- 7.3 Resistance: (4)

$\varepsilon = V_1 + V_2$ $9 = 5,74 + V_2$ $V_2 = 3,26 \text{ V} \checkmark$ $R = \frac{V}{I}$ $R_L = \frac{3,26}{0,168} \checkmark (\text{sub})$ $R_L = \boxed{19,4 \Omega} \checkmark$	OR	$R = \frac{V}{I} \checkmark$ $R = \frac{\varepsilon - V_1}{I_t - \frac{\varepsilon - V_1}{R_1}}$ $R = \frac{9 - 5,74 \checkmark}{0,82 - \frac{3,26}{5} \checkmark}$ $R = \frac{3,26}{0,168}$ $R_L = \boxed{19,4 \Omega} \checkmark$
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- 7.4 Current: (2)

$$I = \frac{V}{R_t} = \frac{\varepsilon}{R_1 + R_2}$$

$$I = \frac{9}{14} = \boxed{0,75 \text{ A}} \checkmark$$

- 7.5 Decrease ✓ (1)

QUESTION 8

16 MARKS

8.1 The total energy supplied per coulomb of charge by the cell ✓✓

(2)

8.2 13V ✓

(1)

8.3

8.3.1 Current:

(2)

$$V_{ext} = \frac{I}{R_{ext}}$$

$$10,5 = \frac{I}{5,6} \quad \checkmark$$

$$I = 1,88 \text{ A} \quad \checkmark$$

8.3.2 Internal:

(3)

$$\varepsilon = V_{ext} + V_{int}$$

$$13 = 10,5 \checkmark + 1,88r \checkmark \text{ (coe: 8.2.1)}$$

$$r = \frac{2,5}{1,88}$$

$$r = 1,31 \Omega \quad \checkmark$$

8.4

8.4.1 DECREASE ✓

(1)

8.4.2 The current through the cell has increased as the resistance in the circuit decreased. ✓

$$I = \frac{V}{R} \quad I \propto \frac{1}{R}$$

The internal voltage increases due to increase in current. ($V \propto I$) ✓Thus the external voltage must decrease as the sum of internal and external voltage will equal the emf. ($emf = V_{ext} + I_{int}$) ✓

(3)

8.4.3 R

(4)

$$emf = V_{ext} + Ir$$

$$13 = IR + (3)(1,31)$$

$$9,07 = 3R$$

$$R = 3,02 \Omega \quad \checkmark$$

$$\frac{1}{R_p} = \frac{1}{R} + \frac{1}{2 \checkmark R_x}$$

$$\frac{1}{3,02} = \frac{1}{5,6} + \frac{1}{2R_x} \quad \checkmark$$

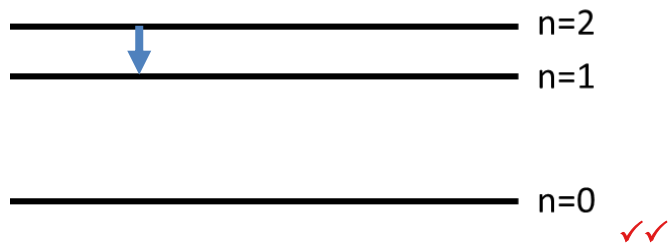
$$\frac{1}{3,02} - \frac{1}{5,6} = \frac{1}{2R_x}$$

$$0,152 = \frac{1}{2R_x}$$

$$R_x = 3,28 \Omega \quad \checkmark$$

QUESTION 9**8 MARKS**9.1 $n=3 \rightarrow n=1$ ✓✓**(2)**

9.2 longest wave

(2)

9.3 energy

(4)

$$E = \frac{hc}{\lambda} \quad \checkmark$$

$$E = \frac{(6,6 \times 10^{-34})(3 \times 10^8)}{618 \times 10^{-9}} \quad \checkmark$$

$$E = 3,02 \times 10^{-19} J \quad \checkmark$$

$$\therefore E = \frac{3,02 \times 10^{-19}}{1,6 \times 10^{-19}} = \boxed{2 \text{ eV}} \quad \checkmark$$